

Statistical data and its types

Where data comes from, how it is grouped, and the four pictures that make it readable.

LESSON 65–67

3 × 40 MIN

ALGEBRA 11, §4.3

IGCSE E20.1–20.2

LESSON CLOCK

18'

26'

30'

28'

30'

3

Kinds of data	18 min
Grouping	26 min
Histograms	30 min
Cumulative frequency	28 min
Practice	30 min
Homework	3 min

LEARNING OBJECTIVES

- Classify data as qualitative or quantitative, discrete or continuous.
- Build a grouped frequency table with sensible classes.
- Draw a histogram with equal and with unequal class widths.
- Draw and read a cumulative frequency curve.

TEXTBOOK

National	pp. 303–320
Cambridge	Core & Extended, pp. 380–396
Workbook	IGCSE Exercise 20.1

01

Explanation

Kinds of data

KIND	MEANING	EXAMPLE
qualitative	a category, not a number	favourite subject
quantitative discrete	counted; only certain values	number of children
quantitative continuous	measured; any value in a range	height, time

POPULATION AND SAMPLE

The **population** is everyone you want to know about; the **sample** is who you actually measure. A sample is useful only if it is **representative** — asking the football team about sport is biased, however many you ask.

Sample size reduces random error; it does nothing at all about bias. That distinction is the single most useful idea in the whole of statistics.

Grouping

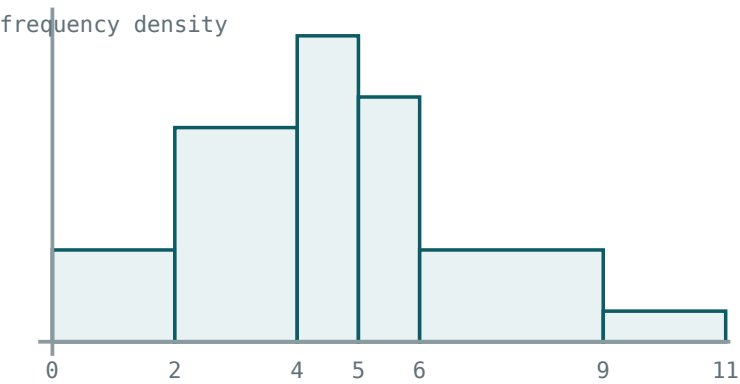
Continuous data must be grouped into **classes**. Three rules make a table readable:

- 1. Between 5 and 12 classes — fewer hides the shape, more shows only noise.
- 2. Classes must not overlap and must leave no gaps: $10 \leq x < 20$, $20 \leq x < 30$.
- 3. Equal widths where possible; unequal only when the data demands it.

GROUPING LOSES INFORMATION

Once heights are recorded as “160–170 cm”, the exact values are gone. Every statistic computed from a grouped table is therefore an **estimate**, and should be described as one.

Histograms



with unequal widths the AREA is the frequency

With unequal widths the vertical axis is frequency density, and the AREA of each bar is the frequency.

$$\text{frequency density} = \frac{\text{frequency}}{\text{class width}}$$

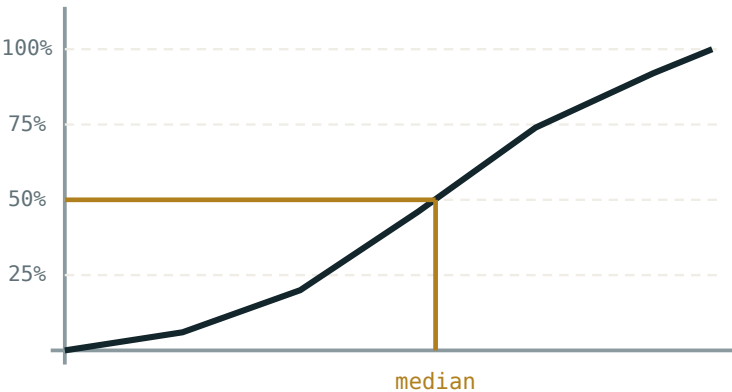
CLASS	FREQUENCY	WIDTH	DENSITY
$0 \leq x < 2$	6	2	3
$2 \leq x < 4$	14	2	7
$4 \leq x < 5$	10	1	10
$5 \leq x < 6$	8	1	8
$6 \leq x < 9$	9	3	3

A HISTOGRAM IS NOT A BAR CHART

Bars touch, because the scale is continuous. And with unequal widths the **height** is not the frequency — the **area** is. Plotting frequency against unequal widths misrepresents the data, which is exactly what the density is for.

Cumulative frequency

Add the frequencies as you go, and plot each running total against the **upper** end of its class. Join the points with a smooth curve.



read across at 50% and down for the median

Read across at 50% for the median, at 25% and 75% for the quartiles.

READ ACROSS AT	READ DOWN FOR
25%	the lower quartile Q_1
50%	the median
75%	the upper quartile Q_3
a given value	how many are below it

UPPER CLASS BOUNDARIES, ALWAYS

A running total of “fewer than 20” is plotted at 20, not at the midpoint. Plotting at midpoints shifts the whole curve and every quartile with it.

02 Worked examples

EXAMPLE 1 A class has 30 in $0 \leq x < 10$ and 30 in $10 \leq x < 40$. Compare the bar heights on a histogram.

① Densities $\frac{30}{10} = 3$ and $\frac{30}{30} = 1$.

② The first bar is three times as tall.

③ But the areas are equal.
Equal frequencies.

ANSWER

Heights 3 and 1; equal areas

EXAMPLE 2 Cumulative frequencies 0, 6, 20, 46, 74, 92, 100 at 0, 2, 4, 6, 8, 10, 11. Estimate the median.

① Total 100; read at 50.

② Between 46 at 6 and 74 at 8.

③ Interpolate: $6 + \frac{4}{28} \times 2 \approx 6.29$.

ANSWER

≈ 6.3

EXAMPLE 3 Why is a survey of a school's football team about sport biased?

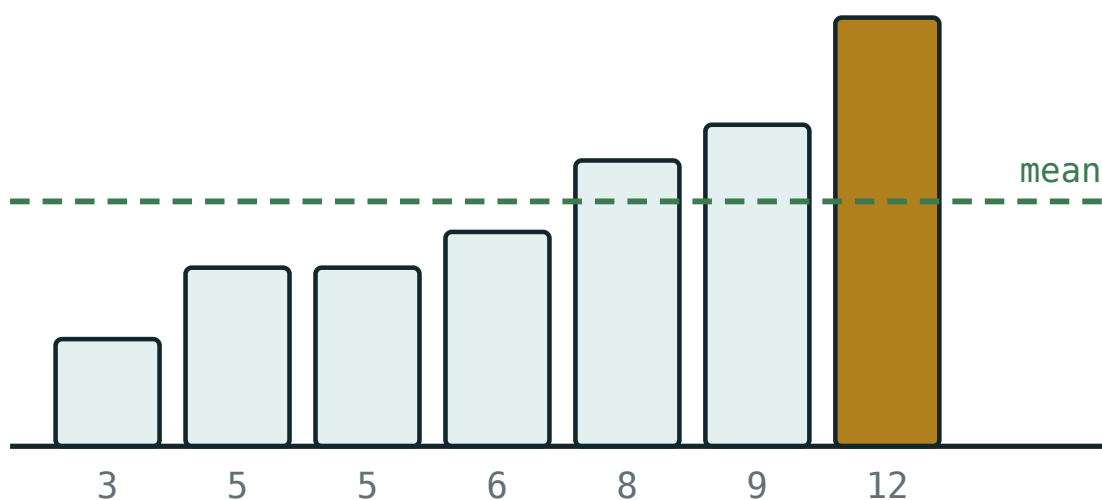
- ① They were chosen for an interest in sport.
- ② They do not represent the school.
- ③ A bigger football team would not help.

ANSWER

The sample is not representative; size does not fix bias

03 Interactive model

Collect one real set of data from the class and group it live on the board.

Grouping and reading data

values 3, 5, 5, 6, 8, 9, 12

mean 6.86

median 6

mode 5

range 9

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Population	Bosh to'plam	Генеральная совокупность
Sample	Tanlanma	Выборка
Qualitative data	Sifat ma'lumot	Качественные данные
Quantitative data	Miqdoriy ma'lumot	Количественные данные
Discrete	Diskret	Дискретные
Continuous	Uzluksiz	Непрерывные
Class interval	Sinf oralig'i	Классовый интервал
Frequency density	Chastota zichligi	Плотность частоты
Cumulative frequency	To'plangan chastota	Накопленная частота
Bias in a sample	Tanlanma xatosi	Смещение выборки

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Height is:

qualitative

discrete

continuous

a category

Number of children is:

qualitative

discrete

continuous

a category

In a histogram with unequal widths the height is:

frequency

frequency density

cumulative frequency

the class width

Cumulative frequency is plotted at:

midpoints

upper class boundaries

lower boundaries

anywhere

A bigger sample fixes:

bias

random error

both

neither

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Is shoe size discrete or continuous?

ANSWER discrete

2. Is time continuous?

ANSWER yes

3. Is eye colour quantitative?

ANSWER no — qualitative

4. Frequency 20, width 4: the density

ANSWER 5

5. Frequency 15, width 5: the density

ANSWER 3

6. Density 6, width 3: the frequency

ANSWER 18

7. Cumulative frequency is plotted at

ANSWER upper class boundaries

Medium · the standard question

7 PROBLEMS

1. 30 in a width-10 class and 30 in a width-30 class: the heights

ANSWER 3 and 1

2. Total 100; the median is read at

ANSWER 50

3. Cumulative 0,6,20,46,74,92,100: the median

ANSWER ≈ 6.3

4. Same data: Q_1

ANSWER ≈ 4.4

5. Same data: Q_3

ANSWER ≈ 8.1

6. Same data: the interquartile range

ANSWER ≈ 3.7

7. Why is a football team a biased sample about sport?

ANSWER Chosen for that very interest

Hard · stretch and reason

7 PROBLEMS

1. A histogram bar has height 4 over $10 \leq x < 25$. Its frequency

ANSWER 60

2. Frequencies 8, 22, 30, 25, 15 over widths 5, 5, 10, 10, 20: the densities

ANSWER 1.6, 4.4, 3, 2.5, 0.75

3. From a cumulative curve, estimate how many are above 7 if the reading there is 60 of 100

ANSWER 40

4. Explain why a grouped mean is an estimate

ANSWER The exact values were lost in grouping

5. Design a sampling method for the heights of a school of 800

ANSWER Stratified by year group, then random within each

6. Why must histogram bars touch?

ANSWER The scale is continuous — no gaps in the data

7. A class of width 1 has density 12. Add a class of width 4 and frequency 12: the new density

ANSWER 3

07 Homework

Homework — 5 tasks

Every histogram must have frequency density on the vertical axis when widths differ.

- Classify five kinds of data you could collect in your school as qualitative, discrete or continuous.
- Group these times (seconds) into five classes and draw the histogram: 12, 15, 18, 19, 22, 23, 24, 25, 27, 28, 30, 31, 33, 36, 40.
- Frequencies 10, 24, 30, 20, 16 over widths 5, 5, 10, 10, 20. Find the frequency densities and draw the histogram.

4. Draw the cumulative frequency curve for the data of task 3 and estimate the median and the quartiles.
5. Explain in three sentences why a larger sample does not correct bias.